

Xiyuan Bao

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[Personal Website](#)

Department of Earth and Planetary Sciences, Harvard University

EDUCATION	Ph.D. in Geophysics , University of California, Los Angeles	2018-Mar 2024
	M.S. in Geophysics , University of California, Los Angeles	2018-2020
	B.S. of Geophysics with Honors , University of Science and Technology of China	2014-2018
	Minor in Computer Science , University of Science and Technology of China	2014-2018
EXPERIENCE	Reginald A. Daly Postdoctoral fellow , Harvard University	Oct 2024-
	Postdoctoral Scholar , University of California, Los Angeles	Mar 2024-Sep 2024
	Graduate Research Assistant , University of California, Los Angeles	Sep 2018-Mar 2024
	<i>Hotspots from Top to Bottom</i> (Advisor: Carolina Lithgow-Bertelloni)	
	Undergraduate Research Assistant , University of Science and Technology of China	Sep-May 2018
	<i>Benchmark on CitcomToEllipsis: Spontaneous subduction towards free surface</i> (Advisor: Wei Leng)	
	Visiting Undergraduate , University of Illinois, Urbana-Champaign	Jun-Aug 2017
	<i>Constraining the Nature of the LLSVPs Using Geoid and Dynamic Topography</i> (Advisor: Lijun Liu)	
RESEARCH INTERESTS	• Dynamics of the Solid Earth Interior • Surface Processes and Coupling with Planetary Interiors • Experimental, Computational and Machine Learning Methods in Geophysics	
GRANTS, HONORS & AWARDS	• Harvard Data Science Initiative-Amazon Web Service (HDSI-AWS) Impact Computing Project (Lead Proposal Writer, PI ineligible due to postdoctoral status, \$209,302) 2026-2028 • Gordon and Betty Moore Foundation Postdoctoral Fellowship (\$192,308) 2026-2028 • Reginald A. Daly Postdoctoral Research Fellowship, Harvard University (\$171,500) 2024-2026 • Study of the Earth's Deep Interior Section Award for Graduate Research, AGU 2023 • Computers and Geosciences Research Scholarship, IAMG 2022 • Jiuzhang Zhao Talent Program in Earth and Space Sciences, USTC 2015-2018 • Award from National Undergraduate Innovation Training Program, USTC 2017-2018 • Earth Science Climbing Scholarship, USTC 2015, 2017 • Silver Scholarship for Outstanding Students, USTC 2016 • First Prize of Physics Innovation Research Experiment Thesis Competition, USTC 2016	
PUBLICATIONS	[7] Lucas, E., Richards, F., Cederberg, G., Bao, X., Hoggard, M. J., Tsuji, S. R. J., Latychev, K., Tsuji, L. J. S. & Mitrovica, J. Emergence of Antarctic mineral resources in a warming world. <i>accepted by Nature Climate Change</i>. [6] Bao, X. & Lithgow-Bertelloni, C. R. Self-correction of the optical distortion effect of thermal plumes in particle image velocimetry. <i>Physics of Fluids</i>. https://doi.org/10.1063/5.0233759 (2024). [5] Bao, X. Giant impact and Earth's mysterious blobs: An interdisciplinary revelation. <i>Science Bulletin</i> 69, 293–294 (2024). [4] Bao, X., Mittal, T. & Lithgow-Bertelloni, C. R. Determining Mid-Ocean Ridge Geography from Upper Mantle Temperature. <i>Earth and Planetary Science Letters</i> 641, 118823 (2024).	

- [3] Zhang, X., Brown, E. L., Zhang, J., Lin, J., **Bao, X.** & Sager, W. W. Magmatism of Shatsky Rise controlled by plume-ridge interactions. *Nature Geoscience*. <https://www.nature.com/articles/s41561-023-01286-0> (2023).
- [2] Li, S., Li, J., Ferrand, T. P., Zhou, T., Lv, M., Xi, Z., Maguire, R., Han, G., Li, J., **Bao, X.**, *et al.* Deep geophysical anomalies beneath the Changbaishan Volcano. *Journal of Geophysical Research: Solid Earth*, e2022JB025671. <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2022JB025671> (2023).
- [1] **Bao, X.**, Lithgow-Bertelloni, C. R., Jackson, M. G. & Romanowicz, B. On the relative temperatures of Earth's volcanic hotspots and mid-ocean ridges. *Science* **375**, 57–61. ISSN: 0036-8075. <https://www.science.org/doi/10.1126/science.abj8944> (2022).

MANUSCRIPTS
SUBMITTED

- [4] Cai, R., **Bao, X.**, Foley, S. F., Giuliani, A., Jiang, S., Pearson, D. G. & Liu, J. *Evidence for a highly fusible fundamental constituent in the convective mantle*. Under review at *Proceedings of the National Academy of Sciences*.
- [3] **Bao, X.**, Wamba, M. D. & Stracke, A. *A deep-mantle geochemical zoning map beneath East Africa and Indian Ocean*. Revision under review at *Science*.
- [2] **Bao, X.** & Lithgow-Bertelloni, C. R. *A method for measuring and analyzing four-dimensional flow fields viscous in Rayleigh-Bénard Convection* submitted.
- [1] **Bao, X.** & Lithgow-Bertelloni, C. R. *Morphology, interactions, and evolution of laminar thermal plumes revealed by four-dimensional velocity measurements and analysis: Application to Earth's mantle* submitted.

MANUSCRIPTS
IN PREP

- [4] Coulson, S., Lloyd, A., **Bao, X.**, Mitrovica, J. X., Dangendorf, S., Pan, L., Valencic, N., Tamisea, M. E., Al-Attar, D. & Heathcotte, D. *Inverting Sea Surface Height Data for Greenland Ice Mass Changes (1993-2019): A Proof of Concept*
- [3] **Bao, X.**, Coulson, S., Valencic, N., Mitrovica, J., Dangendorf, S., Mousavi, M. & Lloyd, A. *Inverting Sea Level Fingerprints for Ice Mass Changes Using Neural Operators: How Much Can We Recover and How Unique Is It?*
- [2] **Bao, X.**, Cao, Z., Gourley, K. C., Gutierrez, S., Liang, Y., Zhou, D., Buffett, B. A., Mittal, T., Li, J., Pierrehumbert, R., Schaefer, L., Orman, J. V. & Foley, B. J. *Coupled Evolution of Earth's Hydrogen Distribution and Thermal History*.
- [1] Wang, W., Zhang, X., Lin, J., Cheng, Z., **Bao, X.**, Zhang, F., Kelley, K. A., Chen, L. & Chu, F. *Ultra-depleted mantle supporting the Marion Rise: implications for ocean rise dynamics*

ORAL

PRESENTATIONS

- [18] **Invited:** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Geodynamics Seminar (Lamont-Doherty Earth Observatory, Palisades, NY, US, Feb. 2026).
- [17] *Inverting Sea Level Fingerprints for Ice Mass Changes Using Neural Operators: How Much Can We Recover and How Unique Is It?* AGU Fall Meeting 2025 (New Orleans, LA, US, Dec. 2025).

- [16] **Invited:** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Brownbag Seminar (Princeton University, Princeton, NJ, US, Nov. 2025).
- [15] **Invited:** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Geophysics Hour (Arizona State University, Tempe, AZ, US, Nov. 2025).
- [14] **Invited:** *Plumes over LLSVPs: Supercharging multi-scale flow resolution with hybrid experimental-adjoint digital twins* Geophysics Seminar (Brown University, RI, US).
- [13] *Plumes over LLSVPs: Supercharging multi-scale flow resolution with hybrid experimental-adjoint digital twins* BiSEPPS Seminar (Harvard University, MA, US, Nov. 2024).
- [12] **Invited:** *A Song of Ice and Fire: Operator Learning for Viscous Flow from Mantle Convection to Ice Dynamics* Daly Special Seminar (Harvard University, MA, US, Feb. 2024).
- [11] **Invited:** *Hotspots from Top to Bottom* Geophysics Seminar (University of Science and Technology of China, Anhui, China, Jan. 2024).
- [10] **Invited:** *Illuminating Mantle Convection: Unraveling Plume Dynamics over LLSVPs in the Laboratory* AGU Fall Meeting 2023 (San Francisco, CA, US, Dec. 2023).
- [9] **Invited:** *Hotspots from Top to Bottom* Earth Sciences Seminar (University of Southern California, CA, US, Oct. 2023).
- [8] *The plume zoo of thermal LLSVPs* AGU Fall Meeting 2022 (Chicago, IL, US, Dec. 2022), V26A–02.
- [7] *Coupled Evolution of Earth’s Hydrogen Distribution and Thermal History* AGU Fall Meeting 2022 (Chicago, IL, US, Dec. 2022), MR33A–06.
- [6] **Invited:** *Hotspots from Top to Bottom* Seismo Lab Seminar (California Institute of Technology, CA, US, Nov. 2022).
- [5] **Invited:** *Are hotspots hotter than ridges?* IGCP 648 Virtual Seminar Series (Curtin University, Australia, Nov. 2022), Series 5, Seminar 5. <https://www.youtube.com/watch?v=30B21c7NGIY>.
- [4] *Experimental Investigation of Purely Thermal LLSVPs* UCLA EPSS Seismology and Tectonics Seminar (UCLA, CA, US, May 2022).
- [3] *Accurate Prediction of Ocean Basins Using Upper Mantle Potential Temperatures* AGU Fall Meeting 2021 (New Orleans, LA, US, Dec. 2021), DI34A–05.
- [2] *Accurate Prediction of Ocean Basins Using Upper Mantle Potential Temperatures* UCLA EPSS Geology/Geophysics Seminar (UCLA, CA, US, Oct. 2021).
- [1] *Correlation of geochemical signals with MORB and OIB temperatures* UCLA EPSS Geocheminar (UCLA, CA, US, Oct. 2019).

POSTER

- PRESENTATIONS
- [13] **Bao, X.,** Wamba, M. D. & Stracke, A. in. AGU Fall Meeting 2025 (New Orleans, LA, US, Dec. 2025).
 - [12] **Bao, X.,** Coulson, S., Valencic, N., Mitrovica, J., Dangendorf, S., Mousavi, M. & Lloyd, A. in. 2025 Gordon Research Seminar: Interior of the Earth (June 2025).

- [11] **Bao, X.** & Lithgow-Bertelloni, C. R. in. AGU Fall Meeting 2024 (Washington DC, US, Dec. 2024).
- [10] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2024 Ada Lovelace Workshop on Modelling Mantle and Lithosphere Dynamics (University of California, Berkeley, CA, US, Sept. 2024).
- [9] **Bao, X.** in. AGU Fall Meeting 2023 (San Francisco, CA, US, Dec. 2023).
- [8] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2023 Gordon Research Seminar: Interior of the Earth (Mount Holyoke College, MA, US, June 2023).
- [7] **Bao, X.** & Lithgow-Bertelloni, C. R. in. Seismo Lab Centennial (California Institute of Technology, US, Nov. 2022).
- [6] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2022 Ada Lovelace Workshop on Numerical Modelling of Mantle and Lithosphere Dynamics (Hévíz, Hungary, Aug. 2022).
- [5] **Bao, X.** & Lithgow-Bertelloni, C. R. in. CIDER2022 (University of California, Berkeley, CA, US, July 2022).
- [4] **Bao, X.,** Lithgow-Bertelloni, C. R. & Jackson, M. G. in. AGU Fall Meeting 2020 (San Francisco, CA, US, Dec. 2020), DI007–0006.
- [3] Wang, Y., Liu, L. & **Bao, X.** in. AGU Fall Meeting 2020 (San Francisco, CA, US, Dec. 2020), DI005–0005.
- [2] **Bao, X.,** Lithgow-Bertelloni, C. R. & Jackson, M. G. in. AGU Fall Meeting 2019 (San Francisco, CA, US, Dec. 2019), DI41D–0028.
- [1] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2019 Gordon Research Seminar: Interior of the Earth (Mount Holyoke College, MA, US, June 2019).

PROFESSIONAL SERVICE

- **Guest Editor**, *PNAS Nexus*, First Call for Papers, in *Machine Learning and Geosciences*
- **Primary Session Convener and Chair**, *Advances in Machine Learning for Solid Earth Geoscience*, AGU Fall Meeting 2025
- **Primary Session Convener and Chair**, *Advances in Deep Earth–Surface Interactions*, AGU Fall Meeting 2024
- **Discussion Leader**, *The Structure and Composition of the Lower Mantle*, Gordon Research Seminar 2023
- **Primary Session Convener and Chair**, *Advances in Mantle Convection and Planetary Evolution*, AGU Fall Meeting 2022
- **Reviewer**, *Nature Geoscience*; *Journal of Geophysical Research: Solid Earth*; *Geophysical Journal International*; *Planetary Science Journal*; *Physics of Fluids*; *Earth and Planetary Physics*; *Frontiers in Earth Science*
- **Reviewer**, National Science Foundation, Geophysics Program

OTHER SERVICE

- Committee Member of UCLA EPSS Family Mentorship Program (EFMP)

OUTREACH

- *Rheology, Plate Tectonics with Food* Saturday Science Academy, UCLA/Charles R. Drew University
Mar 2023
- *Why Hawaii is a volcano? How do Alaskan volcanos differ?* EYU, UCLA
Nov 2022
- **Media Interview** about our *cold hotspots*, e.g. *ScienceNews*, *New Scientist*^{1,2} *Popular Science*, and *Deutschlandfunk*
Jan 2022
- **Invited Outreach Article** *Rocks beneath volcanic hotspots can be surprisingly cool*
Jan 2022
- *The sweet smell of Earth's mantle: Why is Hawaii a volcano?* EYU, UCLA
Nov 2020, 2021

	• Lab visit from Tokyo Tech. UCLA	Feb 2020
	• <i>Why is Hawaii a volcano?</i> Explore Your Universe, UCLA	Nov 2019
TEACHING EXPERIENCE	• <i>Blue Planet: Introduction to Oceanography</i>, Teaching Assistant, UCLA • <i>Earthquakes</i>, Teaching Assistant, UCLA	Spring 2020 Spring 2019, Fall 2021
MENTORING EXPERIENCE	• Jade Wight, undergrad, <i>Modeling Mantle Plumes and Their Effect on Dynamic Topography</i>, won 2020 <i>Deane Oberste-Lehn Scholarship</i>, UCLA • Wan Ki (Arthur) Lo, undergrad, <i>Dynamics of Adjacent Mantle Plumes through Physical and Computer Simulations</i>, won 2021 <i>Harold and Mayla Sullwold Scholarship</i>, UCLA • Kavya Agarwal, undergrad, <i>Compositional and thermodynamical effect on mantle plumes; Critical plume merge distance with shadowgraph</i>, IIT	Oct 2019 - Jun 2021 Jan 2020 - Jun 2022 Sep 2021 - Aug 2022